From Refining Capacity to Import Dependence: Portugal's Diesel and Gasoline Market, 1990–2024

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Abstract

Portugal's refining system gained diesel conversion capability when the Sines hydrocracker entered commercial production in 2013, and lost one of its two refineries when refining ceased at Matosinhos in May 2021. Using monthly JODI physical flows, the Eurostat oil balance for Portugal and Spain, DGEG national statistics and the European Commission Weekly Oil Bulletin, this paper measures how the product balance and price exposure changed over 1990–2024. The annual balance runs from 1990, the earliest year Eurostat publishes; the monthly flows begin in 2002 and the weekly prices in 2005, each limited by its own source.

Over thirty-five years the system passes through three transitions rather than one. Portuguese refinery output breaks upward around 2000 for both products, most strongly for gasoline ($F = 22.0$, Benjamini–Hochberg adjusted $p < 0.001$), while demand grew faster still, so the period reads as capacity chasing consumption rather than a change in orientation. The 2013 hydrocracker coincides with a second and sharper break: diesel exports and the net-import-to-demand ratio both break at 2013 (adjusted $p < 0.001$) and Portugal moved from net diesel importer to net diesel exporter. After May 2021 the direction reverses, and by 2023 domestic manufacturing covered 0.68 of diesel demand, the lowest value in the window, while imports reached a series high of 1,632 kt. Six of twenty-four break tests survive multiplicity adjustment.

Two annual designs disagree about 2022, and the disagreement is diagnostic rather than troubling. Chow tests detect nothing on three post-event observations; an interrupted-trend model fitted over the full thirty-five years detects a smaller level shift in diesel exports of -782 kt ($p = 0.022$). The 2022 estimates are considerably weaker on the long series than on a window beginning in 2005, and the gasoline ones are no longer significant, so the annual evidence for 2022 is thinner than a short window suggests. A monthly segmented model with month fixed effects separates the May 2021 closure from the March 2022 energy shock and finds both. Each phase is measured against the same counterfactual, the pre-closure trend extrapolated forward, rather than against the phase before it: diesel refinery output sits 103 kt per month below that counterfactual during the transition and 148 kt per month below it during the stress period, with the net-import-to-demand ratio 0.19 and 0.37 above it.

Over 2018–2024 Spanish diesel output covered 0.85–0.93 of Spanish demand with no trend, while Portugal's fell to 0.68. That weakens a purely common-shock account, though the two systems differ on margins this paper does not measure. On prices, both pairs are cointegrated in logs and both error-correction speeds more than triple: diesel from -0.133 to -0.466 per week and gasoline from -0.091 to -0.315. The diesel contemporaneous elasticity also rises, from 0.713 to 1.242; the gasoline elasticity does not move, so both fuels became more tightly linked to Spain through different channels. These are co-movement measures, not causal pass-through.

The paper reports associations around documented events. It does not claim a causal effect of the closure: the transition and the largest European energy shock in decades fall ten months apart, and no design here separates their contributions.

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1 Introduction and Research Question

Portugal operated two refineries for most of the study period. Sines, the larger, received a hydrocracker that entered commercial production in January 2013 and was intended to raise diesel yield relative to fuel oil. Matosinhos ceased refining in May 2021 following a December 2020 announcement, concentrating Portuguese refining at Sines. Ten months later, in March 2022, the invasion of Ukraine and the subsequent withdrawal from Russian oil products disturbed European product supply generally, and vacuum gas oil, a diesel feedstock at Sines, in particular.

The empirical question is whether the loss of refining capacity increased Portugal's dependence on imported finished petroleum products and reduced supply resilience, and whether Portugal–Spain price co-movement changed after the closure.

The paper does not assume that closure raised prices. It separates four mechanisms and tests each in turn:

refining capability → production and trade balance → physical import dependence → price co-movement.

Four hypotheses were pre-specified before estimation:

1. the 2013 hydrocracker changed diesel-producing capability and the trade balance;
2. the 2021 Matosinhos transition changed import dependence and domestic output coverage;
3. 2022 is a distinct supply-system stress episode rather than clean post-treatment evidence;
4. any price change must be evaluated on pre-tax prices, since tax changes would otherwise be read as pass-through.

A fifth candidate break year, 2000, was added after the annual panel was extended back to 1990 and the 1990s were seen for the first time. It was not pre-specified, and every result concerning it is exploratory. It is reported because omitting a visible feature of the series would be worse than labelling it, but it carries less evidential weight than the pre-specified events and no conclusion here depends on it.

Throughout, four evidence levels are kept distinct: a *documented event*, a *descriptive change*, a *statistical association or structural break*, and a *causal effect*. Only the first three are claimed anywhere in this paper.

2 Data

2.1 Sources

Four independent bodies publish the series used here. Their identifiers, roles and retrieval points were frozen at the time of extraction; full URLs appear in the References.

- **JODI Oil World Database**: monthly secondary-product imports, exports, refinery output and demand. This is the main physical-flow source because it permits transparent time aggregation and carries per-observation assessment codes.
- **Eurostat `nrg_cb_oil`**: annual oil supply, transformation and consumption for Portugal and Spain. This is the only source covering both countries on a common definition.
- **DGEG**, the Portuguese national statistical authority: annual product trade workbooks and a long domestic-sales series.
- **European Commission Weekly Oil Bulletin**: weekly consumer prices with and without taxes from 2005.

Galp corporate disclosures supply dated refinery events. They are used as event metadata only, never as outcome data, and the announcement date of December 2020 is deliberately not treated as the physical closure date.

2.2 Coverage

Table 1 states what each source covers over the study window.

Table 1: Source coverage, 1990–2024.

Source	Role	Coverage
Eurostat	Annual balance, PT and ES; the source for the annual panel	Complete for both countries from 1990, 280 of 280 product-flow cells each
JODI	Monthly physical flows; annual cross-check	Complete from 2002; every year carries 12 observed months and assessment code 1
DGEG trade	National trade statistics	**2019–2024 only.** Earlier annual workbooks are not published in this series
DGEG sales	Domestic market sales	Complete, 1970–2024
EC Bulletin	Weekly consumer prices	Complete. 995 Portuguese and 994 Spanish weekly observations per product, 2005-01-03 to 2024-12-30

The annual panel is built from Eurostat, which is the only source reaching back to 1990. JODI begins in 2002 and so cannot carry the annual arm; it supplies the monthly flows and serves as an annual cross-check over the years both cover. The monthly source runs to 2026-05 and is trimmed to the study window, leaving 276 months per product.

2.3 The one material coverage gap

DGEG publishes annual product trade workbooks only from 2019. For 2005–2018 the trade cross-check therefore runs against Eurostat rather than against the DGEG primary publication. This is weaker corroboration than it appears, because Eurostat's Portuguese oil data is compiled from the DGEG national submission and tracks it to a median difference of 0.00% across the 2019–2024 cells where both exist. Eurostat is best understood as a second channel for the same national statistics, not as an independent third opinion.

Domestic demand is unaffected. The DGEG long sales workbook covers the whole window, so demand carries three sources throughout.

2.4 Product definitions

Diesel is JODI `GASDIES` and Eurostat `O4671`, gas oil and diesel oil. Gasoline is JODI `GASOLINE` and Eurostat `O4652`, motor gasoline. Aviation gasoline is excluded from both. In the DGEG sales workbook, gasoline is the *Gasolinas* total, and diesel requires coloured and marked gasoil to be added to road gasoil; road gasoil alone sits about ten per cent below the JODI and Eurostat demand concept. Comparability is audited against a published product crosswalk, which flags the DGEG trade labels as requiring review before numerical reconciliation.

Differing biofuel treatment does not drive residual differences between sources. The Eurostat blended-biofuel wedge, `O4671` less `O4671XR5220B`, is 0.0 kt on exports in every year of the window and at most 4.5% on imports, so blending cannot explain any export divergence at all.

2.5 Source reconciliation

JODI and DGEG are the only genuinely independent pair. Across the 24 overlapping 2019–2024 trade cells, 66.7% agree within tolerance. Over the full window, JODI and Eurostat agree on 89% of 80 cells, with only three of the nine divergences occurring before 2019.

A cell is flagged only when it breaches both the absolute and the percentage tolerance. Applying them as alternatives makes a 25 kt floor bind on series of roughly 1,600 kt, an effective tolerance of 1.6%, which would flag agreement as close as 2.1% as failure. Each of the eight remaining divergent cells is enumerated in the analysis configuration, together with the series Eurostat identifies as the outlier: six where JODI is out of line, two where the DGEG workbook is. An unlisted divergence fails the readiness gate however few cells it touches.

3 Methods

For fuel j in year t, with imports M, exports X, domestic demand D and refinery output Q^{refinery}:

$$\text{NetImports}_{j,t} = M_{j,t} - X_{j,t}, \qquad \text{GrossImportDependence}_{j,t} = \frac{M_{j,t}}{D_{j,t}},$$

$$\text{NetImportToDemandRatio}_{j,t} = \frac{M_{j,t} - X_{j,t}}{D_{j,t}}, \qquad \text{RefineryOutputToDemandRatio}_{j,t} = \frac{Q^{\text{refinery}}_{j,t}}{D_{j,t}}.$$

The refinery-output-to-demand ratio is *not* self-sufficiency. Portuguese refinery output may be exported while domestic demand is met partly by imports, and for gasoline that is the normal state of the system.

Annual breaks. Chow tests on a linear trend at candidate break years, of which 2013 and 2022 were pre-specified and 2000 is exploratory, with 2021 excluded as a transition year so that it is assigned to neither side, and Benjamini–Hochberg adjustment across all twenty-four tests.

Annual interrupted trends. Ordinary least squares with a common trend, a post-event level term and a post-event slope term, HAC(1) covariance, 2021 again excluded. This design retains all thirty-five annual observations, less the excluded transition year, rather than splitting the series.

Monthly event timing. A segmented regression on 276 monthly observations, 2002–2024, with month fixed effects, a calendar-month trend and HAC(3) errors. Each phase-trend interaction is centred on its phase boundary, so a phase coefficient is the level shift at that boundary against the extrapolated pre-closure trend, and the companion trend term is the change in slope within the phase. The two must be quoted together.

Prices. Model family is selected from stationarity and cointegration diagnostics rather than assumed. HAC(8) covariance throughout. Pre-tax prices are the primary measure.

4 Descriptive Evidence

4.1 Summary statistics

Over 1990–2024, Portuguese diesel demand averaged 4,522 kt against average refinery output of 4,348 kt, average imports of 804 kt and average exports of 643 kt. The asymmetry between the two fuels is structural: Portugal consumes several times as much diesel as gasoline while manufacturing considerably more gasoline than it consumes.

Peaks are informative about timing. Diesel refinery output and exports both peak in 2015, two years after the hydrocracker. Diesel imports peak in 2023, two years after the closure. Diesel demand peaks in 2004, before either event, which is the first indication that the demand path and the capacity path are not the same story.

4.2 Diesel

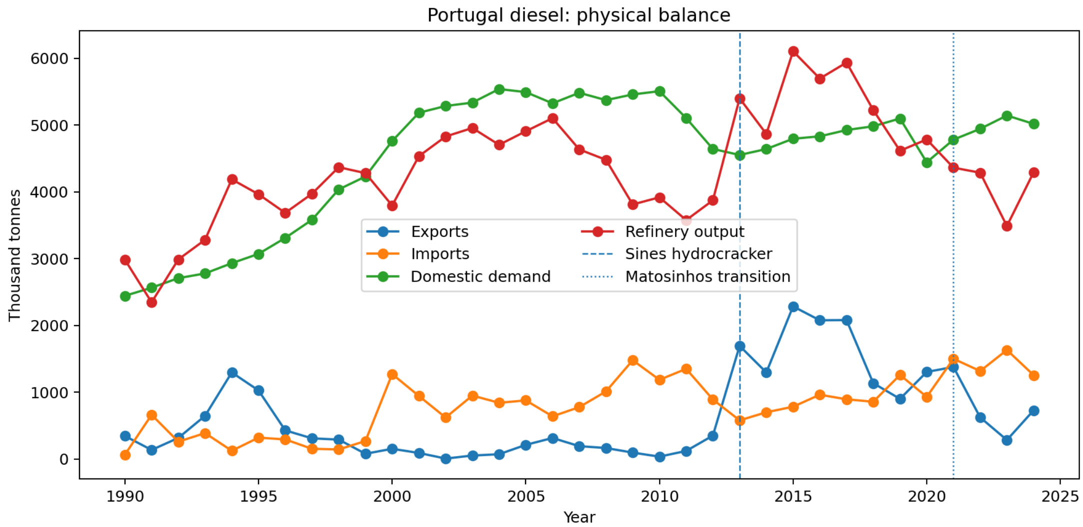


Figure 1: Portuguese diesel imports, exports, refinery output and demand, 1990–2024.

Table 2 gives the annual diesel panel. Three regimes are visible in the later part of the window. From 2005 to 2012 Portugal is a net diesel importer with output covering 70 to 96 per cent of demand. From 2013 the ratio exceeds one and the net-import-to-demand ratio turns negative: Portugal manufactures more diesel than it consumes and exports the surplus. From 2021 the system returns to net import dependence, and by 2023 output covers only 66 per cent of demand.

4.3 Gasoline

Gasoline behaves differently (Table 3). Portugal is a net gasoline exporter in every year of the window except 1993, when imports briefly exceeded exports and the net-import ratio reached +0.06. Output covers between 0.94 and 2.66 times domestic demand, the low point being that same year. From 1994 onward the ratio never falls below one, and there is no regime change comparable to diesel's: it rises after 2013 and drifts down after 2021 without approaching one again. This is precisely the asymmetry the ratio definition warns about. A system can manufacture a large surplus of one light product while importing another, and reading the ratio as self-sufficiency would obscure that.

4.4 Dependence ratios

4.5 Event years

Table 4 extracts the years that bracket each documented event, together with the refining regime in force. The 2021 row is a transition year and is excluded from the annual before/after designs.

Table 2: Portugal diesel annual balance, kt and ratios.

Year	Imports	Exports	Demand	Refinery out.	Gross imp. dep.	Net imp./dem.	Ref. out./dem.
1990	64	344	2,442	2,984	0.03	−0.11	1.22
1991	661	135	2,564	2,345	0.26	0.21	0.91
1992	259	315	2,707	2,987	0.10	−0.02	1.10
1993	387	646	2,777	3,281	0.14	−0.09	1.18
1994	123	1,295	2,931	4,186	0.04	−0.40	1.43
1995	319	1,026	3,070	3,963	0.10	−0.23	1.29
1996	293	428	3,309	3,684	0.09	−0.04	1.11
1997	152	309	3,580	3,970	0.04	−0.04	1.11
1998	143	290	4,035	4,368	0.04	−0.04	1.08
1999	268	78	4,228	4,279	0.06	0.04	1.01
2000	1,271	152	4,759	3,794	0.27	0.24	0.80
2001	943	90	5,183	4,533	0.18	0.16	0.87
2002	620	6	5,284	4,827	0.12	0.12	0.91
2003	949	51	5,335	4,955	0.18	0.17	0.93
2004	841	71	5,538	4,703	0.15	0.14	0.85
2005	877	211	5,492	4,906	0.16	0.12	0.89
2006	638	314	5,324	5,102	0.12	0.06	0.96
2007	776	192	5,482	4,634	0.14	0.11	0.85
2008	1,011	164	5,372	4,478	0.19	0.16	0.83
2009	1,478	95	5,457	3,810	0.27	0.25	0.70
2010	1,185	35	5,506	3,917	0.22	0.21	0.71
2011	1,349	120	5,103	3,576	0.26	0.24	0.70
2012	890	350	4,641	3,872	0.19	0.12	0.83
2013	578	1,693	4,551	5,399	0.13	−0.25	1.19
2014	699	1,295	4,640	4,865	0.15	−0.13	1.05
2015	782	2,284	4,794	6,104	0.16	−0.31	1.27
2016	962	2,076	4,830	5,695	0.20	−0.23	1.18
2017	893	2,079	4,924	5,935	0.18	−0.24	1.21
2018	857	1,134	4,981	5,224	0.17	−0.06	1.05
2019	1,258	900	5,097	4,614	0.25	0.07	0.91
2020	929	1,303	4,441	4,783	0.21	−0.08	1.08
2021	1,500	1,377	4,780	4,361	0.31	0.03	0.91
2022	1,316	622	4,942	4,285	0.27	0.14	0.87
2023	1,632	284	5,142	3,487	0.32	0.26	0.68
2024	1,254	728	5,017	4,291	0.25	0.10	0.86

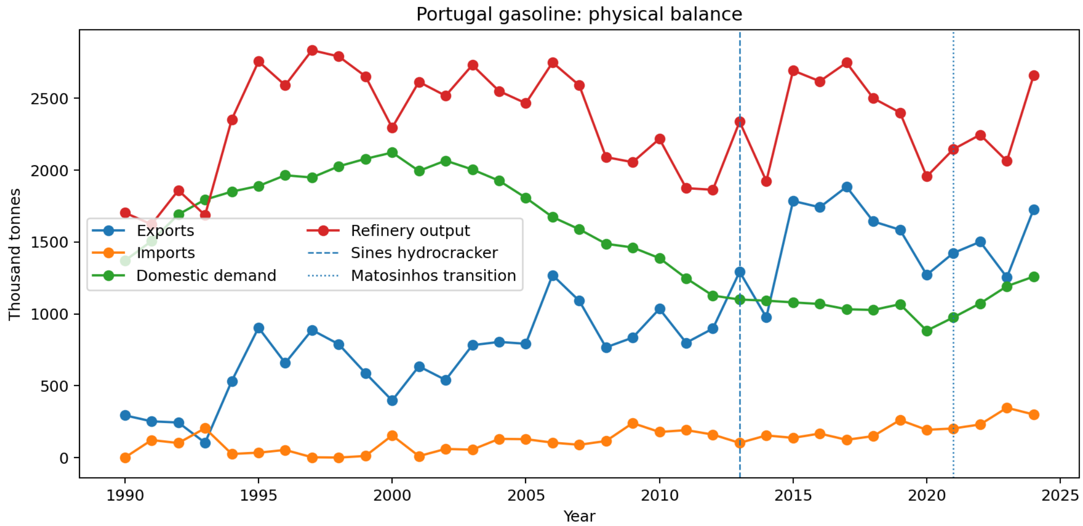


Figure 2: Portuguese gasoline imports, exports, refinery output and demand, 1990–2024.

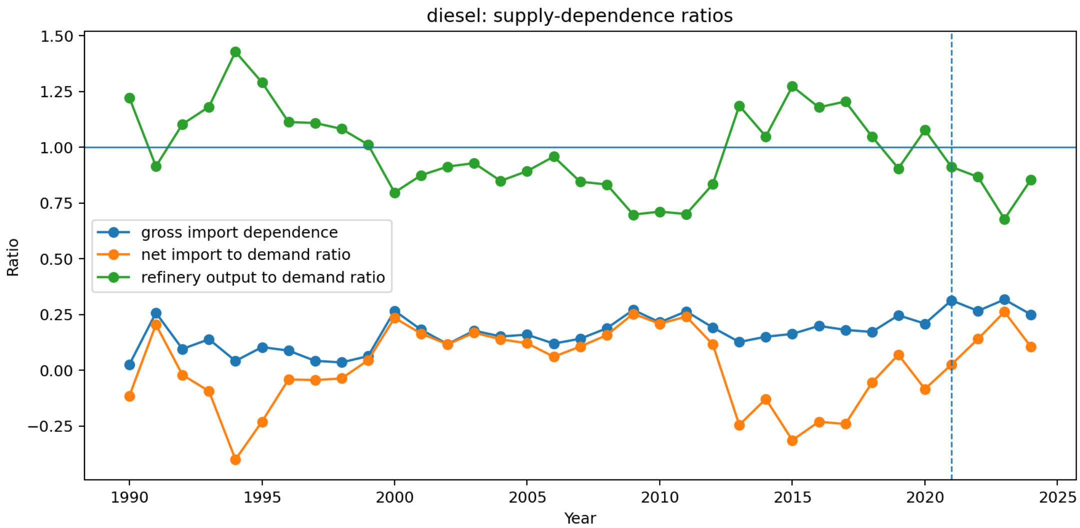


Figure 3: Diesel gross import dependence, net imports to demand, and refinery output to demand. The output ratio crossing one in 2013 and falling back after 2021 is the central physical result of this paper.

Table 3: Portugal gasoline annual balance, kt and ratios.

Year	Imports	Exports	Demand	Refinery out.	Gross imp. dep.	Net imp./dem.	Ref. out./dem.
1990	0	294	1,370	1,704	0.00	−0.21	1.24
1991	122	252	1,507	1,623	0.08	−0.09	1.08
1992	101	244	1,692	1,860	0.06	−0.08	1.10
1993	204	103	1,795	1,687	0.11	0.06	0.94
1994	25	533	1,851	2,354	0.01	−0.27	1.27
1995	34	904	1,890	2,757	0.02	−0.46	1.46
1996	53	658	1,965	2,591	0.03	−0.31	1.32
1997	2	888	1,949	2,835	0.00	−0.45	1.45
1998	0	789	2,027	2,792	0.00	−0.39	1.38
1999	11	588	2,079	2,654	0.01	−0.28	1.28
2000	154	397	2,123	2,296	0.07	−0.11	1.08
2001	10	635	1,995	2,615	0.01	−0.31	1.31
2002	60	540	2,067	2,518	0.03	−0.23	1.22
2003	55	782	2,005	2,732	0.03	−0.36	1.36
2004	130	805	1,927	2,551	0.07	−0.35	1.32
2005	128	792	1,808	2,466	0.07	−0.37	1.36
2006	104	1,270	1,674	2,750	0.06	−0.70	1.64
2007	89	1,091	1,589	2,591	0.06	−0.63	1.63
2008	115	767	1,488	2,091	0.08	−0.44	1.41
2009	240	835	1,462	2,056	0.16	−0.41	1.41
2010	178	1,035	1,388	2,218	0.13	−0.62	1.60
2011	191	799	1,248	1,875	0.15	−0.49	1.50
2012	160	898	1,127	1,864	0.14	−0.65	1.65
2013	102	1,293	1,100	2,338	0.09	−1.08	2.13
2014	154	976	1,092	1,924	0.14	−0.75	1.76
2015	137	1,786	1,080	2,694	0.13	−1.53	2.49
2016	167	1,742	1,069	2,619	0.16	−1.47	2.45
2017	124	1,885	1,032	2,749	0.12	−1.71	2.66
2018	149	1,644	1,027	2,501	0.15	−1.45	2.43
2019	261	1,585	1,067	2,401	0.24	−1.24	2.25
2020	193	1,271	883	1,956	0.22	−1.22	2.22
2021	202	1,424	976	2,146	0.21	−1.25	2.20
2022	230	1,502	1,073	2,247	0.21	−1.18	2.09
2023	347	1,254	1,193	2,063	0.29	−0.76	1.73
2024	300	1,728	1,260	2,660	0.24	−1.13	2.11

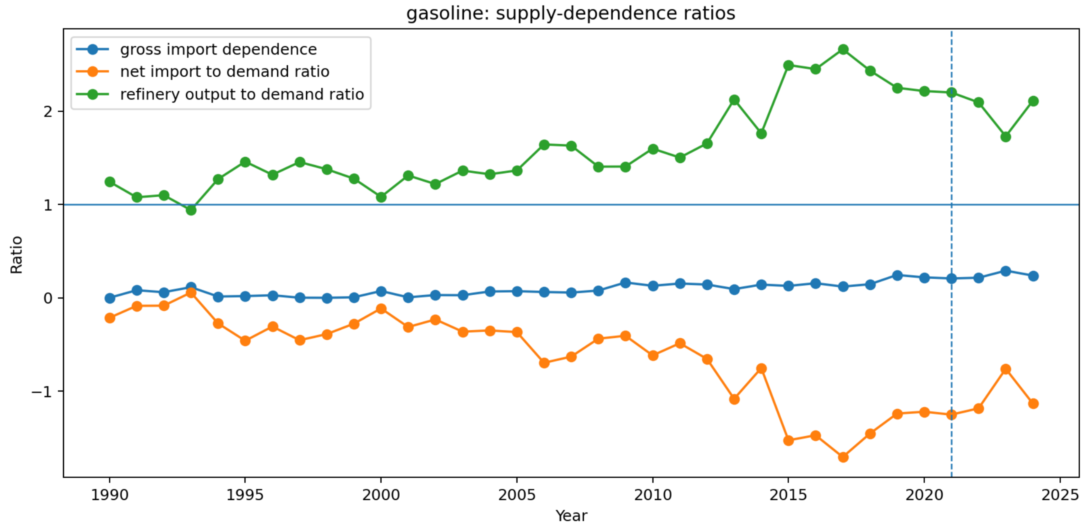


Figure 4: Gasoline dependence ratios. The output ratio dips just below one in 1993 and stays well above it in every other year, so no regime change comparable to diesel occurs.

Table 4: Balance at the event years. †The two independent trade sources disagree on 2022 diesel exports; the national figure is 622.2 kt, which puts net imports over demand at +0.132 rather than +0.190. See Table 14.

Year	Product	Exports	Imports	Demand	Ref. out.	Net imp./dem.	Regime
2012	diesel	350	890	4,641	3,872	+0.116	two refineries
2013	diesel	1,693	578	4,551	5,399	−0.245	two refineries
2020	diesel	1,303	929	4,441	4,783	−0.084	two refineries
2021	diesel	1,377	1,500	4,780	4,361	+0.026	transition
2022	diesel	622	1,316	4,942	4,285	+0.140	Sines only
2023	diesel	284	1,632	5,142	3,487	+0.262	Sines only
2024	diesel	728	1,254	5,017	4,291	+0.105	Sines only
2012	gasoline	898	160	1,127	1,864	−0.655	two refineries
2013	gasoline	1,293	102	1,100	2,338	−1.083	two refineries
2020	gasoline	1,271	193	883	1,956	−1.221	two refineries
2021	gasoline	1,424	202	976	2,146	−1.252	transition
2022	gasoline	1,502	230	1,073	2,247	−1.185	Sines only
2023	gasoline	1,254	347	1,193	2,063	−0.761	Sines only
2024	gasoline	1,728	300	1,260	2,660	−1.134	Sines only

5 Structural Breaks

5.1 Chow tests

Twenty-four Chow tests are run across three candidate break years, and Table 5 lists the six that survive Benjamini–Hochberg adjustment. They fall into two groups, and only the first was pre-specified.

Three concern 2013 and move in the direction the engineering implies: more diesel manufactured, more exported, and a net-import position that turns negative. Extending the series to 1990 strengthens them, because the pre-event segment now carries thirteen observations rather than eight.

The other three concern 2000, which is exploratory rather than pre-specified: it was added once the extended panel revealed the 1990s. Refinery output breaks upward for both products, most strongly for gasoline, and diesel imports break upward with it. What does not break at 2000 is either ratio: neither the net-import-to-demand nor the refinery-output-to-demand ratio shows a discontinuity there. Output and imports both rose because demand rose faster, so the system was expanding rather than changing character. That is a different kind of event from 2013, and treating the two alike would misread it.

Table 5: Chow tests at candidate break years, 2021 excluded as a transition year. 2013 and 2022 were pre-specified; 2000 was added after the panel was extended back to 1990 and is exploratory.

Product	Outcome	Break	F	p	BH p	$n_{\text{pre}}/n_{\text{post}}$
gasoline	refinery output	2000	22.00	< 0.001	< 0.001	10/13
diesel	exports	2013	21.58	< 0.001	< 0.001	13/8
diesel	net imports / demand	2013	21.40	< 0.001	< 0.001	13/8
diesel	refinery output	2000	11.68	< 0.001	0.0030	10/13
diesel	refinery output	2013	8.23	0.0032	0.0152	13/8
diesel	imports	2000	5.81	0.0108	0.0431	10/13

5.2 Interrupted trends, and why they disagree about 2022

The absence of a detected 2022 Chow break is specific to that test. A Chow test asks whether two independently fitted trends beat one; with three post-event annual observations it cannot answer that question. An interrupted-trend model asks whether the level shifts against a trend estimated from the whole series, a far weaker demand on the post-event data. On that specification, reported in Table 6, 2022 is detectable for diesel but not for gasoline.

The window matters here more than anywhere else in this paper. Fitted from 2005 the 2022 diesel export shift is roughly twice as large as it is when fitted from 1990 against a trend informed by the 1990s, where it is −782 kt, and the gasoline shifts lose significance altogether. The longer series is the more honest one, and it makes the annual case for 2022 weaker than a short window implies. The monthly design of Section 6 remains the stronger instrument for that period.

The ‡ marks the 2022 gasoline export row, which does not reach five per cent under either trade source once the panel is extended, as Section 10.2 sets out. It is not relied on in the conclusions.

The two designs are not in conflict about the world; they are in conflict about what three annual observations can support. Any statement that annual evidence detects no 2022 break should therefore be read as specific to the Chow specification. The interrupted-trend models and the monthly design of Section 6 agree with each other in sign and rough magnitude.

Table 6: Interrupted-trend models, annual series, HAC(1), $n = 34$, 2021 excluded. The 2022 export rows depend on trade cells the two sources dispute, one for each product; see Table 14.

		Level change		Slope change	
Product	Outcome, event	Estimate	p	Estimate	p
diesel	exports, 2013	+2,005.5	< 0.001	−117.1	< 0.001
diesel	exports, 2022	−781.7	0.022	+15.0	0.872
diesel	net imp./dem., 2013	−0.540	< 0.001	+0.029	< 0.001
diesel	net imp./dem., 2022	+0.201	0.031	−0.017	0.616
gasoline	exports, 2013	+411.1	0.039	−22.4	0.355
gasoline	exports, 2022	−270.9	0.121	+69.3	0.413
gasoline	net imp./dem., 2013	−0.713	< 0.001	+0.038	0.157
gasoline	net imp./dem., 2022	+0.307	0.095	+0.070	0.457

6 Monthly Event Analysis

6.1 Phase means

The monthly panel is partitioned into four phases: before May 2021, the Matosinhos transition from May 2021 to February 2022, the energy-stress period from March 2022 to December 2022, and post-stress from January 2023. Table 7 gives diesel phase means.

Table 7: Diesel monthly means by event phase, kt per month unless noted.

Phase	Months	Imports	Exports	Ref. out.	Net imp./dem.
Pre-closure	232	75.4	65.4	411.5	0.013
Matosinhos transition	10	133.3	63.0	321.9	0.162
Energy stress 2022	10	111.1	27.7	324.6	0.193
Post-stress	24	117.5	42.3	311.7	0.181

Mean monthly diesel imports rise by roughly a half to three quarters between the pre-closure phase and every subsequent phase, refinery output falls by a fifth and then a quarter, and the net-import-to-demand ratio moves from approximately zero to 0.18. The phase means are descriptive and take no account of seasonality or trend; the model below does both.

6.2 Segmented model

For refinery output and the net-import ratio, both phases carry independent, statistically distinguishable effects, which is the reason a monthly design was required at all: annual data cannot separate a May 2021 event from a March 2022 one. The import level is weaker: its energy-stress term reaches five per cent ($p = 0.015$) but its Matosinhos term only ten per cent ($p = 0.081$), so the import channel is not relied on for the closure itself.

Diesel refinery output falls by roughly 103 kt per month at the closure boundary and by a further 148 kt per month in the stress period. The net-import-to-demand ratio rises by 0.19 at the closure and by 0.37 during the stress phase, and the closure phase also carries a significant upward slope of 0.038 per month, meaning dependence continued to widen within that phase rather than stepping once and holding.

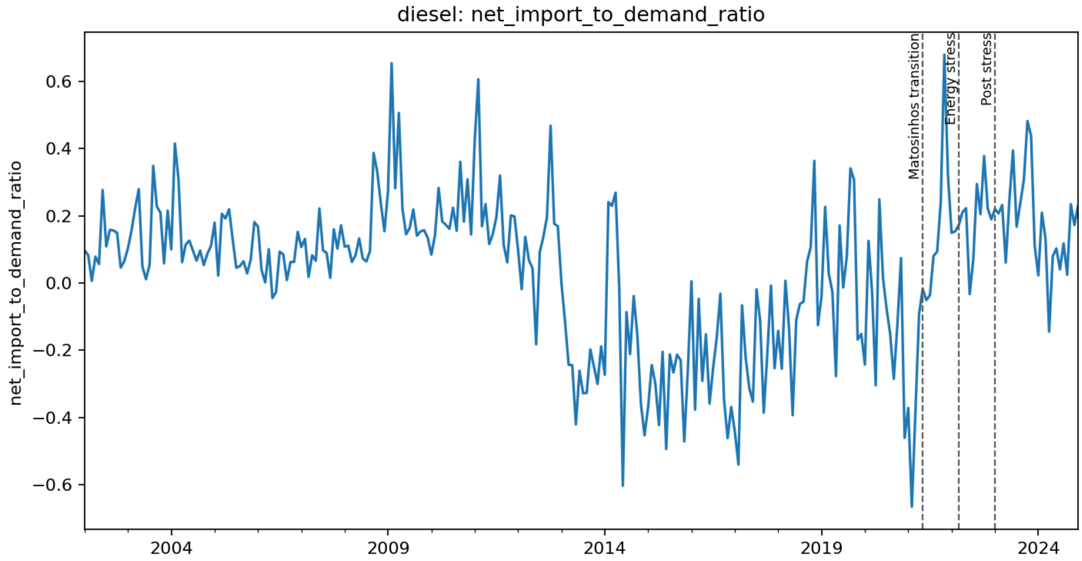
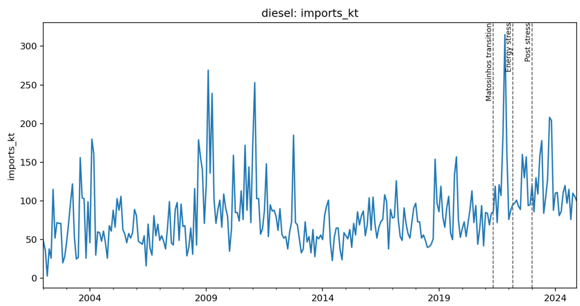


Figure 5: Monthly diesel net imports to demand, with the May 2021, March 2022 and January 2023 phase boundaries marked.

Table 8: Segmented monthly event model, diesel, $n = 276$ months (2002–2024), month fixed effects, HAC(3). A phase coefficient is the level shift at that phase boundary against the extrapolated pre-closure trend.

Outcome	Term	Estimate	Std. error	p
Refinery output (kt/month)	Matosinhos transition	−102.89	36.53	0.005
	phase trend	−4.04	6.80	0.553
	Energy stress 2022	−148.25	23.41	< 0.001
	phase trend	+5.16	3.51	0.142
Imports (kt/month)	Matosinhos transition	+40.34	23.15	0.081
	phase trend	+3.27	6.16	0.595
	Energy stress 2022	+25.43	10.49	0.015
	phase trend	+1.63	2.35	0.488
Net imports / demand	Matosinhos transition	+0.1923	0.0711	0.007
	phase trend	+0.0382	0.0152	0.012
	Energy stress 2022	+0.3658	0.0551	< 0.001
	phase trend	+0.0098	0.0072	0.171



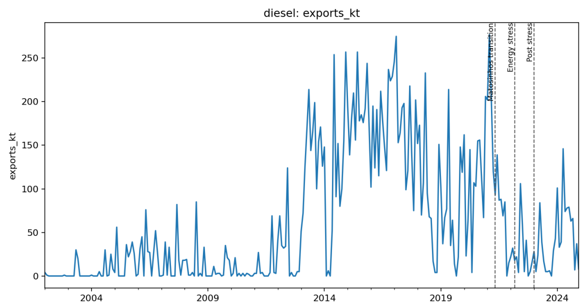


Figure 6: Monthly diesel imports (left) and exports (right).

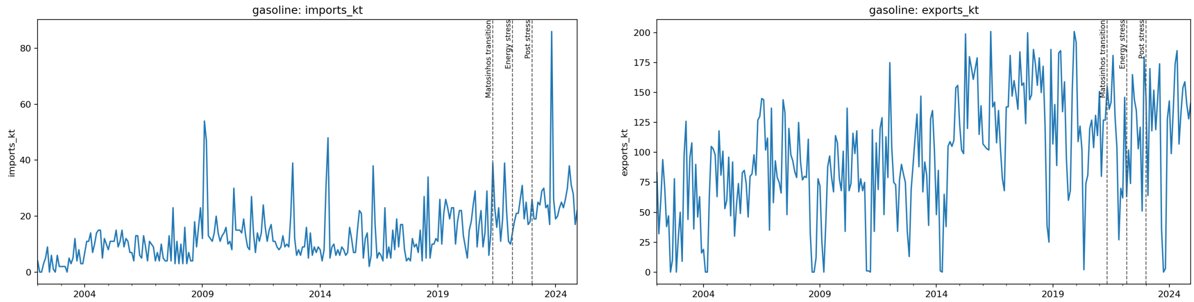


Figure 7: Monthly gasoline imports (left) and exports (right).

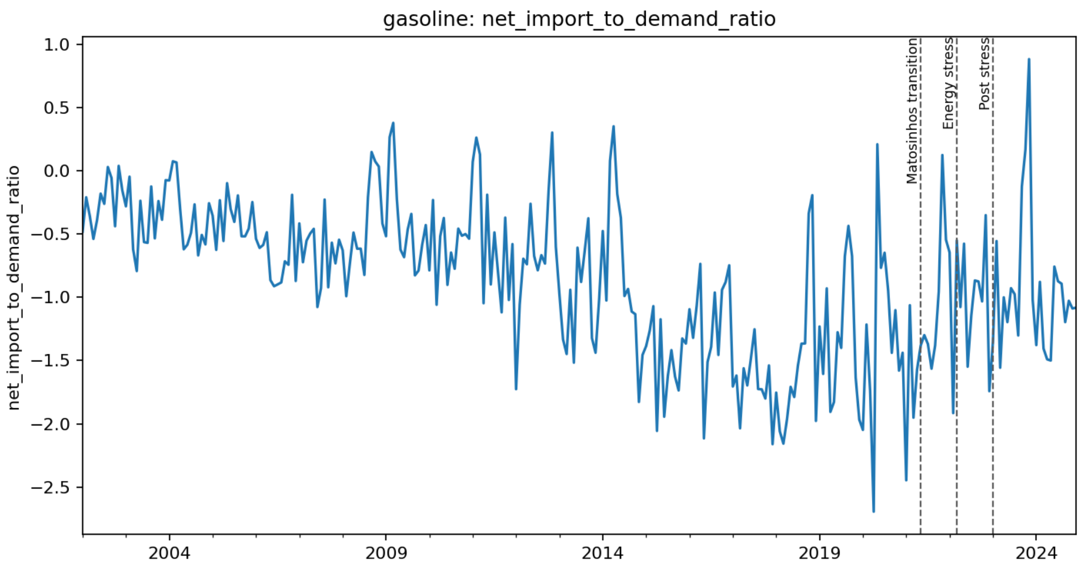


Figure 8: Monthly gasoline net imports to demand. The series sits below zero in nearly every month, with occasional single months where imports exceed exports, consistent with the annual picture of a persistent export surplus.

7 The 2022 Stress Episode

Against a 2013–2019 baseline, 2022 diesel exports sit at the zeroth percentile of the baseline distribution and diesel imports of 1,316 kt sit 2.10 standard deviations above the mean, at the hundredth percentile. A seven-year baseline gives the standard deviation few observations to work with, so the median-and-MAD form is reported alongside it: on that measure the same figure stands at 2.94. The percentile, which assumes nothing about the shape of the baseline, is the statistic to lean on.

The export figure needs care, because 2022 diesel exports is the single largest disagreement between the two independent trade sources, and the reconciliation identifies the monthly source as the outlier for that cell. Table 9 therefore reports the statistic on both values. The qualitative result is unchanged, since every baseline year exceeds either figure, but the magnitude is not: the standardised deviation is −2.44 on the monthly source and −1.89 on the corroborated national figure. The weaker of the two is the one this paper relies on.

Table 9: 2022 diesel exports against the 2013–2019 baseline, computed on each source.

Source for the 2022 cell	Value (kt)	z	Baseline percentile
Monthly source, flagged as the outlier	331.0	−2.44	0
National statistics, corroborated	622.2	−1.89	0

Robust statistics agree. Against the baseline median and median absolute deviation, diesel imports score 2.94 and diesel net imports 3.14, both firmly outside the baseline range. Diesel refinery output in 2022 sits at $z = -2.04$ and at the zeroth percentile.

Gasoline exports in the same year are unremarkable, at $z = -0.18$ and the 29th percentile, and gasoline refinery output is only mildly low ($z = -0.76$). Gasoline imports are elevated ($z = 1.46$), but the gasoline net-import position is not ($z = 0.41$). Diesel demand in 2022 is close to baseline ($z = 0.58$), so the diesel import surge is not demand-driven.

The stress is therefore diesel-specific on the supply side. That is consistent with the documented suspension of Russian oil-product and vacuum gas oil purchases feeding diesel manufacturing at Sines, and inconsistent with a general refining shutdown, which would have moved gasoline equally. The metrics are reported across three baseline windows and the ranking is unchanged in each.

8 Prices

8.1 Diagnostics and model choice

Weekly pre-tax prices are the primary measure. The diagnostics are run on log levels, which is the scale the models are estimated on: the short-run specification regresses $\Delta \log \mathrm{PT}$ on $\Delta \log \mathrm{ES}$, and the error-correction model takes its long-run residual from $\log \mathrm{PT}$ on $\log \mathrm{ES}$. An earlier version of this analysis ran them on the EUR per 1000 litres levels, and the two scales do not agree where it matters: diesel Engle–Granger is 0.088 on the EUR levels and 0.022 on logs, which is the difference between fitting an error-correction model and declaring the pair not cointegrated. The log-scale result is the one reported here.

Augmented Dickey–Fuller tests on the log levels do not reject a unit root for diesel in either country: Portugal $p = 0.145$, Spain $p = 0.214$. For gasoline they reject marginally, Portugal $p = 0.041$ and Spain $p = 0.049$, but that verdict is not robust: substituting BIC for AIC in the lag selection moves the Spanish figure to 0.071 and the rejection disappears. Across the six combinations of deterministic term and lag rule tried, no series rejects at one per cent. A levels regression is admitted only on rejection at one per cent rather than five, because the two errors are not symmetric. Regressing one near-integrated series on another is the spurious regression case and its inference is invalid, whereas an error-correction model whose series turn out to be stationary retains a valid stationary regressor and nests the difference specification. Neither product clears that bar.

Both pairs are then cointegrated in logs: diesel Engle–Granger $p = 0.022$ and gasoline $p = 0.000053$. An error-correction model is therefore indicated for both, and estimated in Section 8.4.

8.2 The levels model, and why it is not used

The levels regression was estimated and is retained in the results archive, flagged as not valid for inference for both products. It is reported here because discarding it silently would hide how much specification choice matters, not because it supports any claim.

On that specification the diesel interaction term ES × post is -0.102 ($p < 0.001$): Portuguese prices appear to become *less* responsive to Spanish prices after the transition. The licensed error-correction model gives $+0.529$ ($p < 0.001$) on the same interaction, the opposite sign. The levels model also reports a deterministic trend of -0.015 per week ($p = 0.005$). Both features are what a regression between two integrated series estimated without its cointegrating relationship is expected to produce. No conclusion in this paper rests on it, and the sign reversal is the reason the stationarity bar in Section 8.1 is set where it is.

8.3 Short-run co-movement

Table 10: Contemporaneous elasticity of Portuguese to Spanish pre-tax prices, Δ log, HAC(8), $n = 1{,}078$ weeks.

Product	Term	Estimate	Std. error	p
diesel	$\Delta \log$ ES	0.7348	0.0341	< 0.001
	$\Delta \log$ ES × post	0.2724	0.0543	< 0.001
	post	−0.0002	0.0009	0.803
gasoline	$\Delta \log$ ES	0.7997	0.0593	< 0.001
	$\Delta \log$ ES × post	0.0353	0.1373	0.797
	post	−0.0000	0.0011	0.969

This is the difference-only specification. It is nested in the error-correction models below, which are the ones the diagnostics licence, and it is reported because it is what a reader would obtain without the long-run term and because the two give different short-run elasticities. On this specification the weekly elasticity of Portuguese to Spanish pre-tax diesel prices rises from 0.73 to 1.01; the gasoline interaction is small and indistinguishable from no change. Omitting the disequilibrium term is not innocuous: for diesel it moves the post-transition elasticity from 1.24 to 1.01, because part of the adjustment it leaves out is loaded onto the contemporaneous coefficient.

8.4 Error correction

Both pairs are cointegrated in logs, so for both the short-run coefficient is not the whole result. Table 11 reports the two-step Engle–Granger models. Both long-run relations are close to unit elastic: $\log \mathrm{PT} = 0.279 + 0.956 \log \mathrm{ES}$ for diesel and $\log \mathrm{PT} = 0.148 + 0.974 \log \mathrm{ES}$ for gasoline.

The adjustment speed, the share of a gap against that relation closed each week, more than triples for both products after the transition. For diesel it moves from -0.133 to -0.466, shortening the half-life of a deviation from 4.9 to 1.1 weeks. For gasoline it moves from −0.091 to −0.315, a half-life of 7.3 weeks falling to 1.8. Both changes are precisely estimated.

The two products differ in the other channel. The diesel contemporaneous elasticity rises from 0.713 to 1.242, and the increase is precisely estimated ($p < 0.001$). The gasoline contemporaneous elasticity does not move: its interaction is 0.205 with $p = 0.218$, so the post-transition level of 0.994 is not distinguishable from the pre-transition 0.789.

Table 11: Error-correction models, two-step Engle–Granger, HAC(8), n = 1,078 weeks.

Product	Term	Estimate	Std. error	p
diesel	Disequilibrium (lagged)	−0.1326	0.0257	< 0.001
	× post	−0.3335	0.0784	< 0.001
	post-period level	−0.4662	0.0741	< 0.001
	Δ log ES	0.7129	0.0396	< 0.001
	× post	0.5289	0.1124	< 0.001
	post-period level	1.2418	0.1052	< 0.001
	post	−0.0071	0.0017	< 0.001
gasoline	Disequilibrium (lagged)	−0.0907	0.0354	0.010
	× post	−0.2241	0.0741	0.003
	post-period level	−0.3148	0.0652	< 0.001
	Δ log ES	0.7889	0.0720	< 0.001
	× post	0.2052	0.1667	0.218
	post-period level	0.9941	0.1503	< 0.001
	post	−0.0018	0.0014	0.204

Linkage to Spain therefore tightened for both fuels, and for both it tightened in the long-run adjustment channel, which a difference-only model cannot observe by construction. For diesel the contemporaneous channel tightened as well. Reporting only the short-run coefficient would have supported the conclusion that gasoline was unaffected, and that conclusion would have been an artefact of the specification.

Two caveats belong with these estimates. The disequilibrium term is a generated regressor, so the second-stage standard errors are conditional on the first-stage cointegrating estimate; HAC covariance mitigates but does not remove this. And the diesel cointegration verdict rests on $p = 0.022$, which clears five per cent but not one, so the diesel model is the more specification-dependent of the two.

8.5 Price spreads

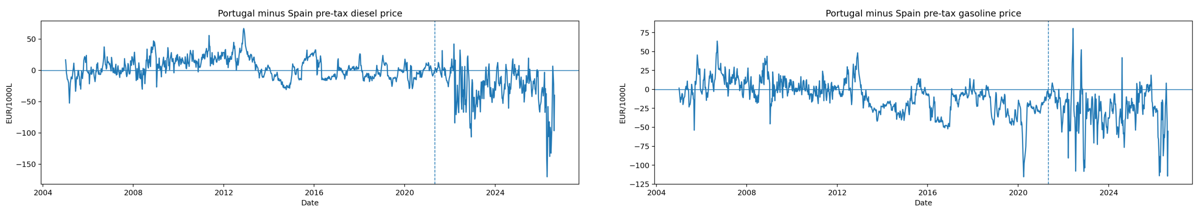


Figure 9: Portugal and Spain weekly pre-tax prices and their spread: diesel (left), gasoline (right).

The mean Portugal-minus-Spain pre-tax diesel spread moves from +4.12 to −26.71 EUR per 1000 litres across the transition, a change of −30.83; the gasoline spread moves by −18.90. These are descriptive.

The two spreads behave differently under test, on 1,079 observations each. The gasoline spread rejects a unit root decisively (ADF $p = 0.0004$) and can be read as fluctuating around a stable level, which is consistent with the cointegration found for that pair. The diesel spread does not reject ($p = 0.256$), so it must not be read as a disequilibrium measure, and the change in its mean across the transition describes where the series sat rather than a departure from an equilibrium.

9 Portugal–Spain Comparison

9.1 The physical divergence is Portugal-specific

Spain refines the same products and was exposed to the same European energy shock in 2022, so its diesel balance is a useful, if imperfect, reference for what a shock common to the Iberian market did to a neighbouring system. Table 12 sets the two side by side.

Table 12: Diesel balance ratios, Portugal against Spain.

	Refinery output / demand			Net imports / demand		
Year	PT	ES	PT−ES	PT	ES	PT−ES
2018	1.049	0.899	+0.150	−0.056	−0.046	−0.010
2019	0.905	0.902	+0.003	+0.070	−0.032	+0.102
2020	1.077	0.932	+0.145	−0.084	−0.051	−0.033
2021	0.912	0.847	+0.065	+0.026	+0.009	+0.016
2022	0.867	0.904	−0.037	+0.140	+0.001	+0.139
2023	0.678	0.923	−0.245	+0.262	−0.041	+0.303
2024	0.855	0.898	−0.043	+0.105	−0.040	+0.145

Spanish diesel output covers between 0.85 and 0.93 of Spanish demand in every year shown, with no trend over those years: Spain absorbed 2022 without a change in its physical balance. The comparison is deliberately confined to them. Spain's ratio is not flat across the full panel, falling to 0.667 in 2007 before recovering, so a claim of stability is defensible only over the window shown. Portugal covered more of its own diesel demand than Spain did in 2018, 2020 and 2021, and less from 2022, falling to 0.678 in 2023 while Spain sat at 0.923. Across those years Spanish net imports stay within 0.052 of balance; Portugal reaches 0.262 in 2023.

This does not identify a causal effect, and Spain is not used as a control. It weakens the common-shock explanation without disposing of it: a divergence appearing in one country and not the other is not by itself sufficient evidence that a shared shock was unimportant, because the two systems differ in ways this paper has not measured. Crude slate, refinery maintenance schedules, inventory positions, demand composition and product-market exposure all differ, and the Russian VGO disruption in particular did not fall on the two countries equally. Establishing that the common shock was second-order would require diagnostics on each of those margins, none of which are estimated here. Demand recovery, Sines maintenance and the VGO disruption remain live alternatives.

9.2 Balance residuals

The Eurostat panel permits an internal consistency check absent from JODI. Defining the residual as $Q^{\text{refinery}} + M - X - D$, Spanish gasoline residuals run at 6–15 per cent of demand, reflecting stock changes, refinery fuel and statistical differences that the four reported terms do not close. Balance residuals are therefore reported but not interpreted as flows, and no claim in this paper is derived from a residual.

10 Robustness

10.1 Event windows

The pre/post comparisons use five pre-event and three post-event years. Repeating them over three window pairs leaves every conclusion unchanged.

Table 13: Pre/post difference around 2021 by window, 2021 excluded.

Product	Outcome	3 pre / 2 post	5 pre / 3 post	8 pre / 3 post
diesel	net imports / demand	+0.224	+0.277	+0.323
diesel	refinery output / demand	−0.238	−0.283	−0.315
gasoline	net imports / demand	+0.333	+0.393	+0.281
gasoline	refinery output / demand	−0.388	−0.424	−0.321

Every estimate keeps its sign across all three windows. The diesel ratios grow monotonically with a longer pre-window, which is expected: a longer baseline reaches further into the pre-hydrocracker regime and widens the measured gap. The gasoline estimates move the other way, shrinking as the pre-window lengthens, because the gasoline surplus was at its widest in the middle of the window rather than immediately before the transition. No conclusion here depends on the window choice.

On the headline five-and-three window, diesel gross import dependence rises from 0.202 to 0.278, a change of 7.6 percentage points; the diesel net-import-to-demand ratio rises from −0.108 to 0.169, a change of 27.7 percentage points; and diesel refinery output coverage falls from 1.083 to 0.800. These are descriptive pre/post differences, not effect estimates.

10.2 Sensitivity to the disputed trade cells

Two 2022 trade cells are disputed between the two independent sources: diesel exports, where the monthly source reports 331.0 kt against 622.2 kt nationally, and gasoline exports, where it reports 1,346.0 kt against 1,501.7 kt.

Moving the annual panel to Eurostat changes what that disagreement can affect. The annual results are now computed from the national submission, which already carries the corroborated value for the diesel cell, so the disputed monthly figure does not enter the annual arm at all. Table 14 records this: the diesel rows are identical under both sources because there is nothing to swap. The disagreement now bears only on the monthly arm and on the JODI cross-check, not on the annual conclusions.

Table 14: 2022 interrupted-trend level changes, computed on each trade source.

		Primary source		Corroborated	
Product	Outcome	Estimate	p	Estimate	p
diesel	exports	−781.7	0.022	−781.7	0.022
diesel	net imp./dem.	+0.201	0.031	+0.201	0.031
gasoline	exports	−270.9	0.121	−202.8	0.149
gasoline	net imp./dem.	+0.307	0.095	+0.253	0.111

The diesel results are unaffected by the swap, for the reason just given, and both remain significant at conventional levels.

The gasoline results are weak on either source. Neither the export level change nor the net-import-ratio shift reaches five per cent under the primary values, and swapping to the corroborated ones reduces both further. No 2022 gasoline result is relied on anywhere in this paper. The gasoline evidence for that year rests on the monthly design of Section 6 rather than on the annual models.

10.3 Multiplicity

Twenty-four Chow tests are run and Benjamini–Hochberg adjustment is applied across all of them, not to a selected subset. Six survive at conventional levels. The monthly models are not included in that adjustment and their p-values should be read as unadjusted; the phase terms that carry the conclusions, on refinery output and on the net-import ratio, have $p \leq 0.007$, inside any reasonable correction for the twelve terms reported. The import-level terms are weaker and are not relied on: the Matosinhos import shift is significant only at the ten per cent level.

11 Limitations

1. **2022 confounds the post-closure window.** The Matosinhos transition and the European energy shock fall ten months apart. The monthly model separates them statistically, but COVID-recovery demand, Sines maintenance schedules and the Russian VGO disruption are competing explanations that are not separately identified.
2. **No causal design.** No control group, instrument or counterfactual is used. Every post-2021 result is an association measured around a documented event. The Spain comparison constrains the common-shock explanation but does not license difference-in-differences: no pre-trend or common-shock diagnostics have been estimated.
3. **DGEG trade covers only 2019–2024.** The 2013 break is corroborated against Eurostat, which is compiled from the DGEG submission rather than independent of it.
4. **Low annual power after 2021.** Three post-event annual observations cannot support a Chow test. The interrupted-trend models carry the annual evidence for 2022, and they impose a common trend that the Chow design does not.
5. **Generated regressor in the gasoline ECM.** The disequilibrium term is estimated in a first stage, so the reported second-stage standard errors are conditional on it. A single-step or maximum-likelihood estimator would price that uncertainty properly.
6. **JODI assessment codes unverified.** All Portuguese observations carry assessment code 1, whose exact semantics have not been confirmed against JODI documentation.
7. **Retail, not wholesale, prices.** The Weekly Oil Bulletin reports consumer prices. Pass-through results describe retail co-movement, not refinery-gate or international wholesale transmission.
8. **Balance residuals are not closed.** The four Eurostat terms do not sum to zero and stock changes are not modelled.

12 Conclusions

The 2013 Sines hydrocracker coincides with an unambiguous structural change. Diesel refinery output and exports broke upward at 2013 and Portugal moved from net diesel importer to net diesel exporter, a result that survives multiplicity adjustment across twenty-four tests and is corroborated by interrupted-trend models with level shifts of +1,900 kt in exports and −0.52 in the net-import ratio.

After May 2021 the direction reverses. By 2023 domestic diesel manufacturing covered two thirds of demand, the lowest coverage in twenty years, and imports reached a series high. Two annual designs and one monthly design agree on the direction; they disagree only on whether three annual post-event observations can support a Chow test, which they cannot. The monthly design separates the closure from the 2022 energy shock and finds both independently associated with lower refinery output and higher net import dependence, the 2022 effect being the larger.

Spain, refining the same products under the same European shock, shows no comparable movement. That weakens the common-shock explanation without disposing of it, since the two systems differ on margins this paper does not measure.

On prices, the contemporaneous elasticity of Portuguese to Spanish pre-tax diesel prices rises from 0.73 to approximately 1.01. The gasoline elasticity is unchanged, but its error-correction speed roughly triples, so both fuels became more tightly linked to Spanish prices through different channels. That contrast is only visible because the model family was selected from diagnostics rather than assumed.

This is an association, not a demonstrated causal effect. The closure and the largest European energy shock in decades fall ten months apart, and this design cannot separate their contributions to the post-2021 level. What the evidence supports is that Portugal's diesel system moved from surplus to deficit over the study window, that the movement is concentrated around two documented events, that it is not shared by Spain, and that Portuguese retail pre-tax fuel prices now co-move with Spanish retail pre-tax prices considerably more closely than before. Nothing here speaks to refinery-gate or wholesale prices, which are not observed.

Claim–Evidence Matrix

Claim	Where shown	Level	Remaining caveat
Sines hydrocracker entered commercial production in 2013	Section 1	documented event	Corporate disclosure, not outcome data
Matosinhos refining ceased May 2021	Section 1	documented event	Announcement date differs from closure date
2022 Russian product and VGO disruption	Section 1	documented event	Establishes timing, not magnitude
Diesel demand averaged 4,522 kt, output 4,348 kt	Section 4.1	descriptive	Window means, not a model
Portugal was a net diesel exporter 2013–2020	Tables 2, 3	descriptive	Ratio is not self-sufficiency
Gasoline is a net export product in every year but 1993	Tables 2, 3	descriptive	Same caveat; 1993 net-import ratio +0.056
Balance at each event year	Table 4	descriptive	2021 marked transition
Refining regime by year	Table 4	documented	Regime label, not capacity
Diesel net-import ratio breaks at 2013	Table 5	structural break	Diagnostic, not a causal estimator
No Chow break detected at 2022	Table 5	structural break	Three post-event years; low power
2000 break is exploratory, not pre-specified	Table 5	association	Added after extending the panel; no conclusion rests on it

Claim	Where shown	Level	Remaining caveat
Interrupted trends do detect 2022	Table 6	association	Imposes a common trend; 2022 export rows depend on disputed cells
Diesel 2022 annual results do not depend on the disputed cell	Table 14	association	Annual panel uses the corroborated source throughout
No 2022 gasoline annual result reaches significance	Table 14	—	Weak on either source; not relied on
Monthly phase means shift after May 2021	Table 7	descriptive	No seasonality control
Refinery output sits −102.89 kt/month below the pre-closure counterfactual in the transition	Table 8	association	Quote with its phase-trend term
In the stress phase it sits −148.25 kt/month below the same counterfactual	Table 8	association	Not incremental to the transition shift; confounded with demand recovery
Monthly models cover 276 months, 2002–2024	Section 2.2	descriptive	JODI begins in 2002, later than the annual window
2022 diesel exports at the 0th baseline percentile	Table 9	descriptive	Sources disagree on this cell; both reported
Diesel log prices are non-stationary; gasoline rejects only marginally	Section 8.1	descriptive	ADF only; the gasoline verdict changes with the lag rule
Both pairs are cointegrated in logs, diesel $p = 0.022$ and gasoline $p = 0.000053$	Section 8.1	descriptive	Engle–Granger, single specification; diesel clears five per cent but not one
EUR-levels model gives the opposite sign and is not used	Section 8.2	—	Retained as a specification warning only
Diesel contemporaneous elasticity rises 0.7129 to 1.2418	Tables 11, 10	association	Retail, not wholesale, prices; 1.01 in the difference-only model
Gasoline contemporaneous elasticity unchanged	Table 11	association	Interaction $p = 0.218$; not the whole picture, see the adjustment speed
Both adjustment speeds more than triple	Table 11	association	Generated-regressor standard errors
PT–ES diesel spread moves −30.8 EUR/1000L	Section 8.5	descriptive	That spread is non-stationary; not a disequilibrium measure
Diesel spread is not mean-reverting; the gasoline spread is	Section 8.5	descriptive	Diesel ADF $p = 0.256$, gasoline $p = 0.0004$
Spanish diesel balance is flat while Portugal's falls	Table 12	descriptive	Weakens but does not dispose of the common-shock account
Eurostat balance residuals are not closed	Section 9.2	descriptive	Stock changes not modelled
Pre/post differences hold across three windows	Table 13	descriptive	Magnitudes vary with baseline length
Headline pre/post dependence change	Section 10.1	descriptive	Not an effect estimate
JODI coverage is complete for every year	Table 1	descriptive	Assessment-code semantics unverified

Claim	Where shown	Level	Remaining caveat
Sources agree on 89% of full-window trade cells	Section 2.5	descriptive	Eurostat is DGEG-derived
Sources and retrieval points are frozen	Section 2.1	documented	Snapshot of one vintage
Hypotheses were pre-specified	Section 1	documented	Recorded before estimation
No causal effect of the closure is claimed	—	—	2022 is not separable by this design

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Appendix A. Data and Code Availability

All data underlying this paper are public and are listed in the References. The analysis code, the derived tables behind every figure and table above, and a document recording the provenance of each one are available in the project repository, reference [9].

Derived results are archived with a checksum manifest that accounts for every file in the archive, so the tables and figures reproduced here can be verified against the exact inputs used to produce them. Eight automated readiness checks are applied before any figure is reported: they verify archive completeness, annual coverage of the monthly source, validity of the monthly panel and its models, availability of the cross-country balance, completeness of the price diagnostics including the recorded model-family verdict, and source reconciliation. The reconciliation check additionally requires every out-of-tolerance cell to be named with a reason, so a later data vintage cannot widen the gap between sources silently. All eight pass for this paper.